LAB 11.5 — Test Suite Summary Document

Module 11: Flutter Mobile Development | **Project:** Taskly App | **Consolidation Status:** 100% Passed

1. Executive Summary

This document summarizes the results of the complete automated test suite engineered for the **Taskly** Flutter application. The test suite spans pure Dart **Unit Tests**, UI **Widget Tests**, route **Navigation Tests**, and end-to-end **Integration Tests**.

13	Total Test Executions
5	Test Suites
100%	Pass Rate
0	Failures / Errors

2. Test Matrix and Verified Behaviors

Test Suite	Test Category	Test Case Description	Result
`task_model_test.dart`	Unit Test	Default completed value is false	**PASSED**
		toggle() switches completed status true ↔ false	**PASSED**
		copyWith() creates updated instance without mutating original	**PASSED**
`task_repository_test.dart`	Unit Test	addTask() adds task to repository and updates count	**PASSED**
		deleteTask() removes target task by ID	**PASSED**
		updateTask() replaces task matching unique ID	**PASSED**

Test Suite	Test Category	Test Case Description	Result
		clear() purges all tasks from collection	**PASSED**
`task_list_widget_test.dart`	Widget Test	Empty State: displays "No tasks yet. Add one!" when empty	**PASSED**
		Add Task: enter text → tap add → verify UI updates	**PASSED**
		Multiple Tasks: add two tasks → verify both visible	**PASSED**
`task_navigation_test.dart`	Navigation Test	Tap seeded task → navigate to TaskDetailScreen with expected title and TextField key	**PASSED**
`task_integration_test.dart`	Integration Test	Full flow: Add "Original title" → Open detail → Edit to "Updated title"	**PASSED**

Test Suite	Test Category	Test Case Description	Result
		→ Save → Verify in list (widget directory)	
		Full flow: Complete lifecycle test execution (integration directory)	**PASSED**

3. Test Architecture & Design Principles

- **AAA Pattern (Arrange → Act → Assert):** Every test file clearly segregates the setup state, action execution, and assertion validation.
- **Explicit Key Finding:** Widgets are targeted with robust `Key` identifiers (e.g. `Key('taskInputField')`, `Key('addTaskButton')`, `Key('detailTitleField')`, `Key('saveTaskButton')`) to prevent brittle UI tests.
- **Asynchronous Pump Control:** Uses `pump()` for immediate frame updates and `pumpAndSettle()` for full routing and transition completion.

4. Known Limitations & Future Roadmap

1. **Persistence:** In-memory repository clears on app restart; future versions will integrate SQLite or Hive.
2. **Offline Sync:** No remote sync mechanism is currently implemented in this module.
3. **Soft Deletes:** Tasks are hard-deleted immediately; future versions can include an undo snackbar.